

Your Article Title

Your Name

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Abstract

Your abstract goes here. This is a brief summary of your article.

1 Introduction

This paper presents the methodology to classify neighbourhoods based on the observed mobility behaviours of their residents. Similar to existing geodemographic classifications (Wyszomierski et al., 2024).

2 Literature Review

3 Methods

Table 1: Variables used in the travel behaviour and demographic clustering analysis. All variables are sourced from the National Travel Survey (NTS) 2024. Diary-derived variables are computed from the 7-day travel diary; self-reported variables are from the interview questionnaire.

Variable	Type	Description
<i>A. Individual demographic variables</i>		
Sex_B01ID	Binary	Sex of respondent (male / female).
Age_B01ID	Ordinal	Age of respondent in 21 banded categories (under 1 to 80+).
MaritalStat_B01ID	Categorical	Legal marital status.
EthGroupTS_B02ID	Binary	Ethnic group. Two categories: White; Non-White.
COB_B01ID	Categorical	Country of birth in 7 categories.
EcoStat_B02ID	Categorical	Economic activity status in 6 categories.
NSSec_B03ID	Ordinal	National Statistics Socio-Economic Classification (NS-SEC) at the 5-category high-level breakdown.
EdAttn3_B01ID	Ordinal	Highest educational qualification attained.
Carer_B01ID	Binary	Whether the respondent provides unpaid care for a family member or friend.
<i>B. Household demographic variables</i>		
HHoldStruct_B02ID	Categorical	Household structure in 6 categories.

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Variable	Type	Description
HHoldNumAdults	Count	Number of adults (aged 18+) in the household.
HHoldNumChildren	Count	Number of children (aged under 18) in the household.
Ten1_B02ID	Categorical	Tenure type in 3 categories.
AddressType_B01ID	Categorical	Dwelling type.
NumCarVan_B02ID	Ordinal	Number of cars or light vans available to the household in 3 banded categories.
HHIncQDS2008_B01ID	Ordinal	Equivalised household income quintile derived from the diary sample, using 2008 bandings.
HHoldGOR_B02ID	Categorical	Government Office Region of the household's residence.
Settlement2021EW_B02ID	Categorical	Urban–rural classification of residence based on the 2021 Census in 3 categories: urban, rural, and other.
<i>C. Modal profile (diary-derived)</i>		
walk_share	Continuous	Proportion of diary trips made on foot, ranging from 0 to 1.
cycle_share	Continuous	Proportion of diary trips made by pedal cycle, ranging from 0 to 1.
car_driver_share	Continuous	Proportion of diary trips as a car or van driver, ranging from 0 to 1.
car_pass_share	Continuous	Proportion of diary trips as a car or van passenger, ranging from 0 to 1.
pt_share	Continuous	Proportion of diary trips by public transport (bus, rail, underground, and taxi), ranging from 0 to 1.
active_share	Continuous	Proportion of diary trips by active modes (walking and cycling combined), ranging from 0 to 1.
main_mode	Categorical	The single mode group accounting for the highest share of diary trips. One of: walk, cycle, car, public transport, or other.
mode_entropy	Continuous	Shannon entropy of the distribution of trips across the five mode groups ($H = -\sum p_i \ln p_i$). Higher values indicate greater modal diversity.
<i>D. Activity profile (diary-derived)</i>		
commute_share	Continuous	Proportion of diary trips made for commuting purposes.
business_share	Continuous	Proportion of diary trips made for employer's business.
education_share	Continuous	Proportion of diary trips for education or escorting to education.
shopping_share	Continuous	Proportion of diary trips for shopping.
escort_share	Continuous	Proportion of diary trips for escorting purposes other than education.

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Variable	Type	Description
personal_share	Continuous	Proportion of diary trips for personal business (e.g. medical appointments, banking).
leisure_share	Continuous	Proportion of diary trips for leisure and recreational activities.
dominant_activity	Categorical	The single trip purpose accounting for the highest share of diary trips.
activity_entropy	Continuous	Shannon entropy of the distribution of trips across the eight purpose categories. Higher values indicate a more varied daily activity programme.
<i>E. Trip characteristics (diary-derived)</i>		
mean_trips_per_day	Continuous	Average number of trips per day, computed as total diary trips divided by 7.
mean_stages_per_trip	Continuous	Average number of stages per trip across the diary week. A proxy for journey complexity and multimodality.
mean_trip_dist_miles	Continuous	Mean trip distance in miles, excluding short walk legs (TripDisExSW).
total_dist_week_miles	Continuous	Total distance travelled across the diary week in miles, including all trip legs.
local_trip_share	Continuous	Proportion of diary trips under 2 miles. Captures the degree of hyperlocal travel behaviour.
long_trip_share	Continuous	Proportion of diary trips over 25 miles. Captures the frequency of longer-distance travel.
peak_share	Continuous	Proportion of trips departing during AM peak (07:00–08:59) or PM peak (16:00–17:59).
weekend_share	Continuous	Proportion of diary trips made on a Saturday or Sunday.
mean_start_time_mins	Continuous	Mean trip departure time expressed in minutes past midnight. Captures habitual early or late travel patterns.
mean_trip_time_mins	Continuous	Mean total trip duration in minutes, including waiting and walking time at each end.
mean_daily_travel_time_mins	Continuous	Mean time spent travelling per day in minutes, computed as total weekly travel time divided by 7.
mean_party_size	Continuous	Mean number of people travelling together per trip, derived from stage-level party size.
solo_travel_share	Continuous	Proportion of trips made alone (party size of one).
<i>F. Self-reported mode use frequencies (interview)</i>		
freq_car	Ordinal	Self-reported frequency of private car use, recoded to a 0–6 numeric scale (0 = never; 6 = three or more days per week).
freq_bus	Ordinal	Self-reported frequency of local bus use (0–6 scale).

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Variable	Type	Description
freq_train	Ordinal	Self-reported frequency of surface rail use (0–6 scale).
freq_walk	Ordinal	Self-reported frequency of walking for 20 minutes or more without stopping (0–6 scale).
freq_cycle	Ordinal	Self-reported frequency of bicycle use (0–6 scale).
freq_taxi	Ordinal	Self-reported frequency of taxi or minicab use (0–6 scale).
freq_phv	Ordinal	Self-reported frequency of app-based private hire vehicle use (e.g. Uber), recoded to the same 0–6 scale.
<i>G. Work from home and digital substitution (interview)</i>		
wfh_any	Binary	Whether the respondent worked from home on at least one day during the diary week (1 = yes; 0 = no).
wfh_frequency	Ordinal	Usual frequency of working from home, recoded to a 0–6 numeric scale (0 = never; 6 = three or more times per week).

3.1

4 Results

Present your results.

4.1 A Subsection

You can add subsections as needed.

5 Discussion

Discuss your findings.

6 Conclusion

Your conclusions.

References

Wyszomierski, J., Longley, P. A., Singleton, A. D., Gale, C., and O'Brien, O. (2024). A neighbourhood output area classification from the 2021 and 2022 UK censuses. *The Geographical Journal*, 190(2):e12550.